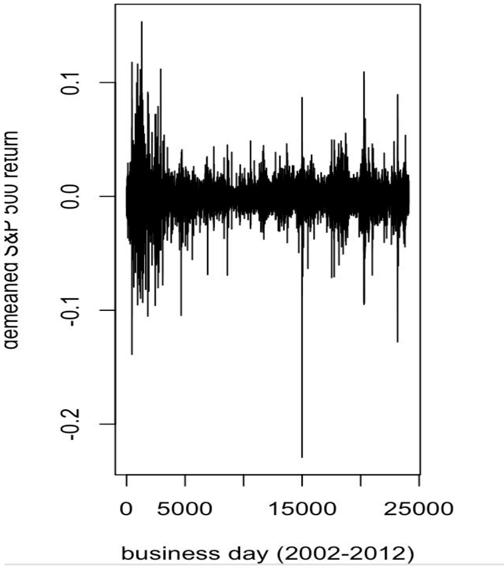
Bitcoin Analysis Graphs

By: Uday Goel

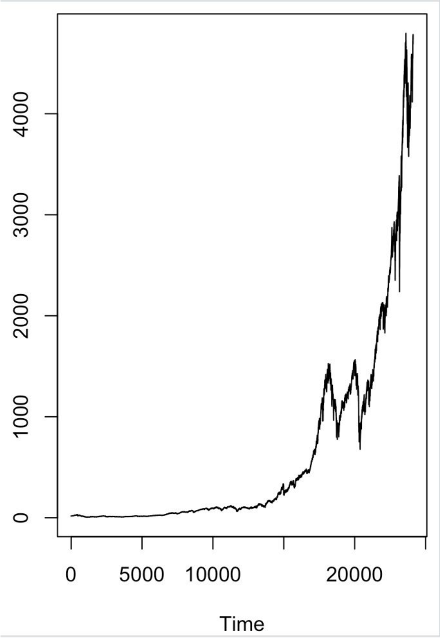
Demeaned Daily Returns of the S&P 500 (2002-2012).

Explanation: This shows the daily returns of the S&P 500, with the mean subtracted ("demeaned"). This makes it comparable to the Bitcoin returns. We can observe that while volatility clustering is present (e.g., around the 2008 financial crisis), it appears visually less extreme than for Bitcoin.



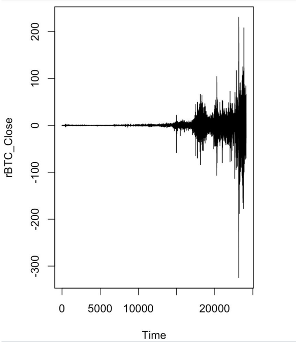
S&P 500 Index Price.

Explanation: This shows the price of a traditional financial asset. Like Bitcoin, it is non-stationary. It serves as a benchmark to contrast with the more volatile cryptocurrency market.

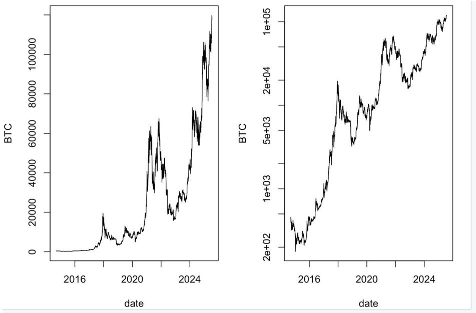


Daily Returns of Bitcoin Over Time.

Explanation: This plot shows the daily returns, which are the day-to-day percentage changes in price. This transformation, known as differencing, is a standard method to achieve stationarity. Unlike the price series, the returns hover around a mean of zero. Most importantly, this graph clearly reveals the core phenomenon for our analysis: volatility clustering. Notice how periods of high fluctuation (e.g., 2018, 2021, 2022) are clumped together, and are distinct from calmer periods. This conditional heteroskedasticity—where the volatility depends on the recent past—is precisely what GARCH models are designed to capture.



2 Graphs



Left: Figure 1: Daily Price of Bitcoin (BTC) from 2016 to 2024.

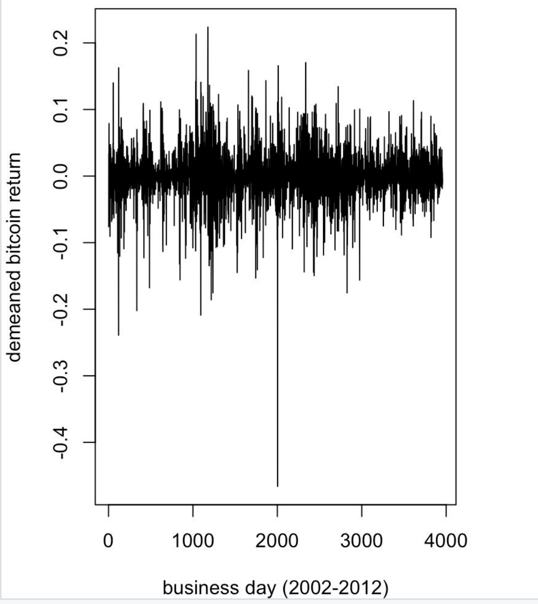
Explanation: This graph shows the raw price of Bitcoin over the last several years. The immediate takeaway is that the series is non-stationary; its mean and variance are not constant over time. The clear upward trend and changing volatility make it impossible to model directly with standard time series techniques that assume stationarity.

Right: Figure 2: Daily Price of Bitcoin (BTC) from 2016 to 2024 (Logarithmic Scale).

Explanation: By plotting the price on a logarithmic y-axis, we are essentially looking at the log-price. This transformation accomplishes two things: first, it helps visualize periods of exponential growth more clearly; second, it stabilizes the variance, making the fluctuations at lower price levels comparable to those at higher levels. However, the series remains non-stationary due to the persistent trend. To properly model this, we must first transform the data to achieve stationarity.

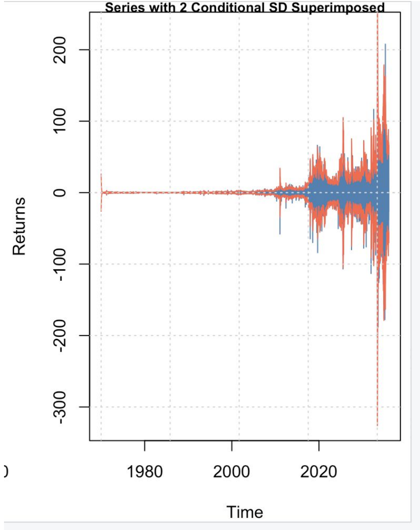
Demeaned Bitcoin Daily Returns (2002-2012).

Explanation: This plot shows demeaned Bitcoin returns from an earlier, and perhaps less mature, period of its history. Comparing this to the more recent data in Figure 3 can reveal how the asset's statistical properties may have evolved over time.



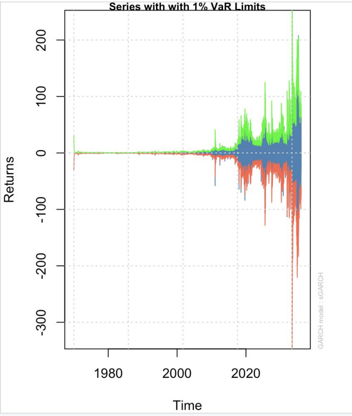
Bitcoin Returns with Superimposed GARCH Conditional Volatility Bands.

Explanation: This plot overlays the estimated conditional volatility from the GARCH model (the red and blue bands, likely representing +/- 2 conditional standard deviations) on top of the actual daily returns. This visually demonstrates the model's fit. In an effective model, the bands should expand during volatile periods to contain the large price swings and contract during calm periods, which appears to be the case here.



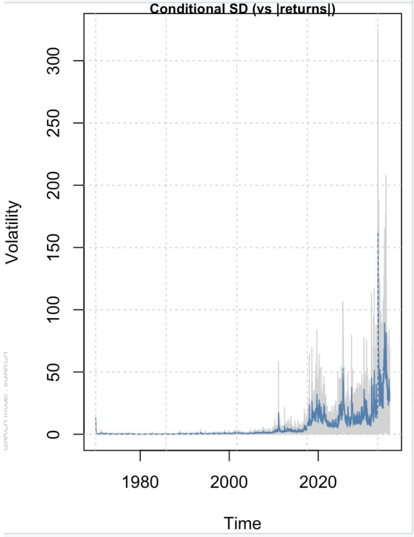
GARCH-Based 1% Value-at-Risk Limits for Bitcoin Returns.

Explanation: This plot demonstrates a key application of GARCH models in financial risk management. Value-at-Risk (VaR) estimates the maximum potential loss over a given period for a given confidence level. Here, the GARCH model's daily volatility forecast is used to calculate the 1% VaR limit. This means that on any given day, we can be 99% confident that the losses will not exceed the boundary shown by the model. The plot shows how this risk measure adapts over time, widening during periods of high volatility.



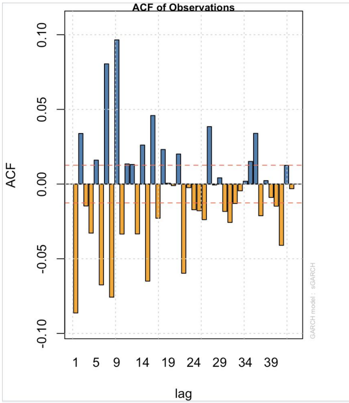
Estimated Conditional Volatility of Bitcoin from the GARCH Model.

Explanation: This graph shows the primary output of the GARCH model: a time-varying estimate of Bitcoin's daily volatility (standard deviation). The model successfully captures the volatility clustering we observed visually in Figure 3. The plot shows clear spikes in estimated volatility that correspond to turbulent periods in the market (e.g., around 2020-2022).



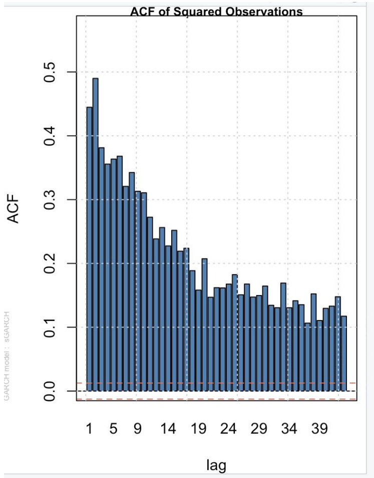
Autocorrelation Function of Daily Bitcoin Returns.

Explanation: The ACF plot shows the correlation of the returns series with its own past values (lags). For Bitcoin returns, we see that nearly all the autocorrelation spikes are within the insignificant range (the dashed blue lines). This lack of significant serial correlation in the returns themselves is consistent with the Efficient Market Hypothesis and confirms that a simple ARMA model for the mean would likely be insufficient.



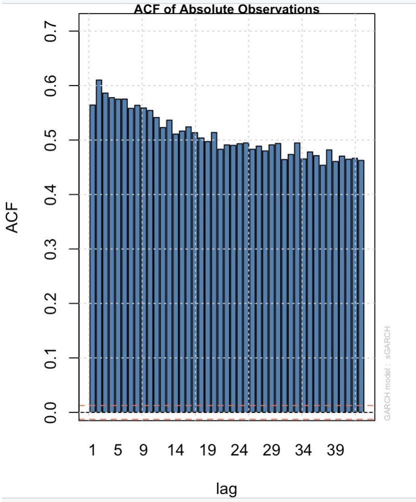
Autocorrelation Function of Squared Daily Bitcoin Returns.

Explanation: This is the most critical plot for justifying our model choice. While the returns are uncorrelated, their squared values (a proxy for variance) show highly significant and slowly decaying autocorrelations. This is the classic statistical signature of ARCH (Autoregressive Conditional Heteroskedasticity) effects, or volatility clustering. It provides definitive evidence that the variance of Bitcoin returns is not constant but is autocorrelated, making a GARCH model appropriate.



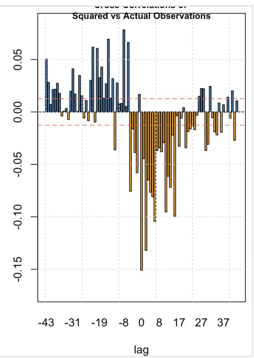
Autocorrelation Function of Absolute Daily Bitcoin Returns.

Explanation: This plot serves as a robustness check for the previous finding. Using absolute returns instead of squared returns also serves as a proxy for volatility but is less sensitive to extreme outliers. The result is the same: significant and persistent autocorrelations, further confirming the presence of volatility clustering and the need for a GARCH model.



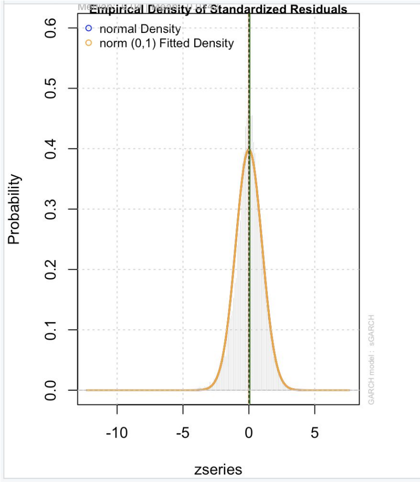
Cross-Correlation Between Squared and Raw Returns to Test for Leverage Effects.

Explanation: This advanced diagnostic tests for the leverage effect, a well-known phenomenon where negative news (bad returns) tends to increase future volatility more than positive news (good returns) of the same magnitude. The CCF plots the correlation between past returns (at various lags) and future squared returns (volatility). An asymmetric pattern, particularly significant negative correlations for past returns, suggests the presence of a leverage effect. This finding could motivate an extension of the project to an asymmetric GARCH model (like EGARCH or GJR-GARCH) for a more accurate representation of the data.



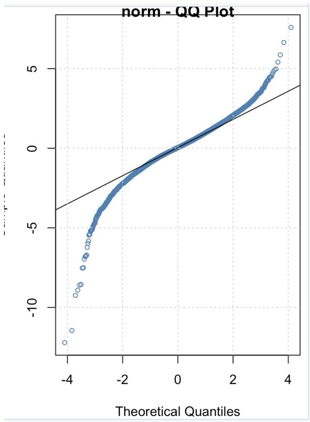
Empirical Density of Standardized GARCH Residuals.

Explanation: This plot confirms the finding from the Q-Q plot. It shows the actual probability distribution of the standardized residuals (in grey) compared to a standard normal distribution (yellow line). The residuals' distribution is more peaked in the center and has heavier tails, confirming they are leptokurtic. This is a common finding in financial data and suggests that for a more refined model, a different error distribution (like the Student's t-distribution) might be more appropriate.



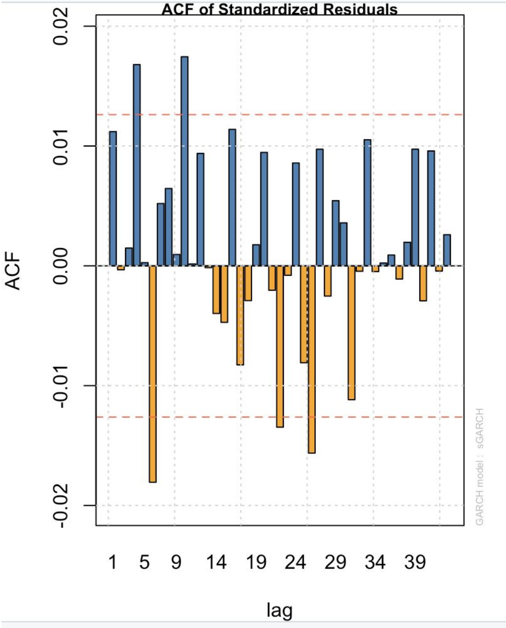
Normal Quantile-Quantile (Q-Q) Plot of Standardized GARCH Residuals.

Explanation: After fitting a GARCH model, we analyze its standardized residuals (the raw residuals divided by the estimated conditional volatility). This Q-Q plot compares the quantiles of these residuals to the quantiles of a theoretical normal distribution. The distinct "S" shape, where the points deviate from the reference line at the tails, is classic evidence that the residuals are not normally distributed. Specifically, they have "fat tails" (leptokurtosis), meaning extreme events are more likely than a normal distribution would suggest.



ACF of Standardized GARCH Residuals.

Explanation: This is a diagnostic check for the model's mean equation. Since there are no significant spikes, it indicates that our model has successfully accounted for any linear dependence (autocorrelation) in the returns.



ACF of Squared Standardized GARCH Residuals.

This is the most important diagnostic plot for a GARCH model. We are checking for remaining ARCH effects after fitting the model. The fact that there are no significant spikes in the ACF of the squared standardized residuals is a sign of success. It demonstrates that our GARCH model has effectively captured the volatility clustering we identified in Figure 5.